

# What's in the Fridge? I Built an App That Remembers.

By Todd Durant · September 2026 · [www.todhdurant.com/blog](http://www.todhdurant.com/blog)

**Abstract.** Twelve thousand years of food preservation, from drying and salting to the Persian ice house and the 1927 electric refrigerator, solved the problem of keeping food and never touched the problem of remembering it. This post follows that gap to its cost today: about \$56 of food a week for a U.S. family of four, roughly 70 million tons a year nationally, more than half of it starting in homes, 41 percent of it ending in landfills where it produces 58 percent of landfill methane. It explains why good food gets thrown out (quality dates read as safety dates), what works against it, and ends with a working prototype, a refrigerator door that keeps its own inventory by voice, and the question its design still needs answered. Every figure is sourced; 28 references follow the text.

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## IN SHORT

- The United States wastes about 70 million tons of food a year, close to a third of the supply (ReFED).
- More than half of that waste starts in homes, and 41 percent of it ends in a landfill.
- Food rotting in landfills makes 58 percent of the methane coming out of them (EPA).
- Every step in 12,000 years of keeping food got better at KEEPING it. None got better at REMEMBERING it. The last section is about the fridge that does.

The EPA says a family of four throws away about \$56 of food a week. That is not a crisis number. It is a grocery-run number. It is the bag you carry in on Saturday and carry out to the bin on Thursday.

This post is about where that bag goes, why it goes there, and the one thing nobody ever built to stop it. I will give you the numbers, and I will give you the source for every one of them. If you only want the short version, watch the video. If you want to know where the numbers come from, keep reading.

*12,000 years of keeping food, and the one problem we never solved*

#### WHAT IT COSTS YOU

**About \$56 of food a week goes in the trash.**

**\$56** household of four  
a week

**\$14** a week, per person  
**\$243** a month, household  
**\$2,913** a year, household  
**11%** of everything you spend on food

Other estimates, per person per year (2023 dollars):



The EPA says its own number is probably LOW: it prices everything at grocery prices and cannot count restaurant food.

Sources: U.S. EPA, Estimating the Cost of Food Waste to American Consumers, April 2025 (2023 data) — \$728/person/yr; household of four = per-person × 4; ~11% of food spending; EPA's literature range \$496–\$1,640 normalized to 2023 dollars · ReFED 2026 (\$762/person) · NRDC 2017 (\$1,800/household).



One hundred grocery dollars. Eleven go in the trash. Chart: Todd Durant. Data: U.S. EPA, Estimating the Cost of Food Waste to American Consumers, April 2025.

## 1 of 12 · Where did keeping food start, and what did the refrigerator change?

People have been trying to keep food from going bad for at least twelve thousand years. The first way was the simplest. You put meat or fish in the sun and the wind, and you let it dry. Dry food does not rot, because the tiny living things that rot it need water.

The first thing we ever added to food to keep it was salt. The oldest hard proof is a pile of fish bones in Sudan, crusted in salt, from about 6,400 years before Christ. Salt pulls water out of food the same way the sun does, only faster. It was valuable enough that the word salary comes from the Latin for salt, though the old story that Roman soldiers were paid in it has no ancient source behind it.

Then came cold. About four thousand years ago, kings in what is now Iraq and Syria dug pits in the ground, lined them with wood and straw, and filled them with winter ice carried down from the mountains. One king, Shulgi of Ur, was so

proud of his that he named a whole year of his reign after it. This is the true ancestor of the box in your kitchen. Britannica puts it plainly: stored ice was the main way people kept food cold until about 1900.

The Persians made it a science. Around 400 BC they built domed mud-brick ice houses in the desert, with a shallow pool beside them that froze on winter nights. They were making ice on purpose, in a place where it never snowed.



A yakhchal, or ice house, in Yazd province, Iran. Persian engineers were storing summer ice in the desert around 400 BC. Photo: reibai, via Wikimedia Commons, CC BY 2.0.

For the next two thousand years the idea did not change much. Ice was cut from lakes in winter, packed in straw, and sold in summer. In the 1800s it moved into the kitchen as the icebox, a wooden cabinet with a block of ice in the top. A man came to your door every few days with a new block.

The electric refrigerator arrived in 1913 and almost nobody bought it. The one that changed everything was the General Electric Monitor-Top in 1927. It sold



more than a million units. After that, keeping food cold was not a chore anymore. It was a hum in the corner of the room.

And that is the problem. Every step in this story got better at keeping food. Not one of them got better at remembering it. The refrigerator keeps food so well, for so long, and so quietly, that we stopped looking. The rest of this post is about what happens when we stop looking.



Cutting ice from a Minnesota lake, around 1870. Blocks like these fed the iceboxes of America for a hundred years. Charles A. Zimmerman, Minnesota Ice Harvest, about 1870. The J. Paul Getty Museum, Los Angeles. Getty Open Content Program, CCo.



Twelve thousand years in one line. Chart: Todd Durant. Sources listed on the chart.

## 2 of 12 · How much food does America waste?

About a third of all the food grown in the United States is never eaten. ReFED, a nonprofit that tracks food waste, put the 2024 number at 70 million tons. That is 29 percent of the whole food supply. It was worth 380 billion dollars. That works out to 412 pounds of surplus food for every person in the country, every year.

It is not an American problem. The European Union measured 58 million tonnes in 2023. The United Nations counted just over a billion tonnes worldwide in 2022. Wherever people have enough to eat, they throw a lot of it away.

I know how a number like a billion tons lands. It lands like nothing. It is too big to picture, so the brain files it under things that are not my fault. If that is you, keep reading, because the next chart is the one that changed how I think about this.

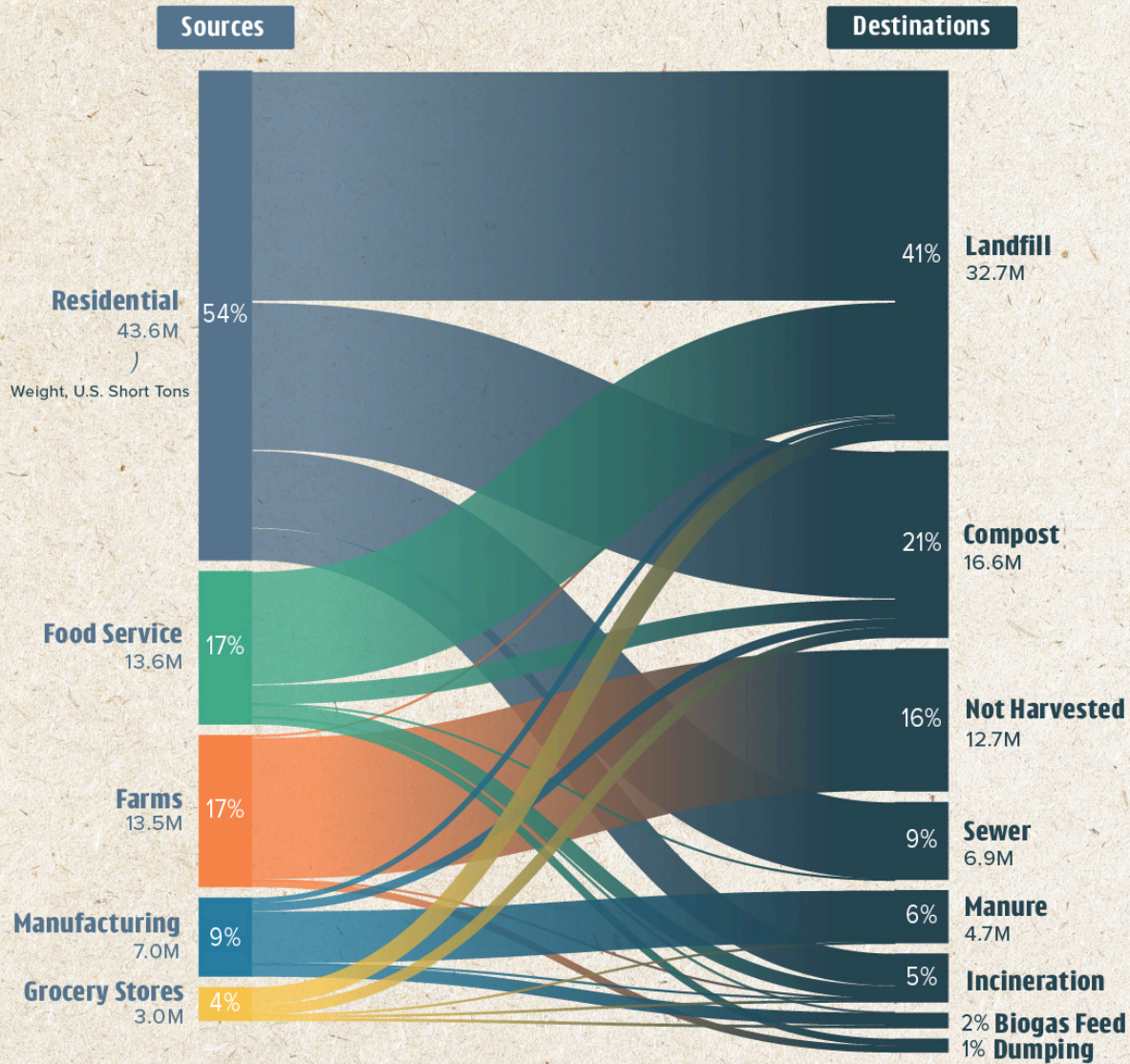


# Food Waste

## IN THE UNITED STATES

There was **80 million U.S. tons** of food waste in the United States in 2021.

Here is where it came from and where it went.



Research & Design: Selin Oguz | Source: ReFED | Numbers may not add up due to rounding.

Where U.S. food waste came from in 2021, and where it went. More than half of it started in homes, and 41 percent of it ended in a landfill. Credit: [Visual Capitalist](#), "[Visualized: Food Waste in the United States](#)" (Selin Oguz, April 2024). Data: ReFED. Used unedited, with attribution, under Visual Capitalist's free-use terms.

## 3 of 12 · Who wastes the most food? Households.

Look at the left side of that chart. The biggest bar is not farms. It is not stores. It is not restaurants. It is households, at 54 percent. More than half of the food wasted in America is wasted after it gets home.

I checked whether that was a fluke. It is not. The European Union measured its own food waste in 2023 and got 53 percent from households. The United Nations counted the whole world and got 60 percent. The United Kingdom got 60 percent too. Four different groups, three continents, one answer.

Here is the part I did not expect. Households are also the only part of the system that is getting better. ReFED found that total food waste in America fell about 2 percent from 2023 to 2024, and almost all of that drop came from homes. Nearly a million tons less. Nobody passed a law. People just started paying a little more attention.

So the biggest single lever in the whole system is not in a factory or a boardroom. It is behind the door in your kitchen.

#### WHERE IT HAPPENS

## More than half of wasted food is thrown away at home — everywhere that measures it.

### United States

ReFED, 2021



### European Union

Eurostat, 2023



### World

UNEP, 2022



### United Kingdom

WRAP



Four groups, three continents, one answer. And in the U.S., homes are the only sector that shrank in 2024 — nearly a million tons less (ReFED 2026).

Sources: ReFED via Visual Capitalist, U.S. food waste 2021 · Eurostat 2025, EU food waste 2023 · UNEP Food Waste Index 2024, world 2022 · WRAP, UK household food waste. Each dataset defines its sectors slightly differently; all four put homes above half.

Four datasets, one answer. Chart: Todd Durant. Sources on the chart.

## 4 of 12 · How much food does the average family waste, in dollars?

Now the money. The numbers below all come from government or university studies, and they do not all agree, so I am going to show you the range instead of picking the one I like best.

The EPA published the newest estimate in April 2025. It works out to \$728 per person per year, or \$2,913 for a household of four. That is \$56 a week, and it is about 11 percent of what a household spends on food. The EPA says this is probably low, because it counted everything at grocery prices even though restaurant food costs more, and it could not count the food you buy away from home at all.

Other estimates are higher. ReFED puts it at about \$762 per person. The Natural Resources Defense Council put it at \$1,800 per household back in 2017, and that is the number you have probably heard. In Britain, the government's waste program says a family of four throws away about a thousand pounds a year. When



the EPA lined up every study and adjusted them all to 2023 dollars, the range ran from \$496 to \$1,640 per person.

One honest caveat. Every one of these household figures is a per-person number multiplied by four. Nobody measured actual four-person families. A real family probably wastes a little less per person than four people living alone, because they share meals. So treat the household number as a ballpark. The per-person number is the one that was actually measured.

If you are not in the United States, the shape is the same. The European Union measured about 130 kilograms (290 pounds) of food waste per person in 2023. The United Kingdom puts it at about a thousand pounds sterling a year for a family of four. Worldwide, about one fifth of the food that reaches shops, restaurants and homes is thrown away. And for readers who think in metric: the 412 pounds of surplus food per American is about 187 kilograms.



The EPA's own graphic for its 2025 cost estimate. U.S. Environmental Protection Agency.

## **5 of 12 · Why do we throw away good food? (Sell by vs. use by, explained)**

Nobody sets out to waste food. So why does it happen? Three reasons, and each one has a number behind it.

We forget. A 2019 survey of 2,000 Americans, paid for by the appliance maker Bosch, found that people throw out about four spoiled items from the fridge every week. Three out of four of them said the reason was simple: they forgot the food was in there, or it got lost behind something else. A 2026 survey by Frigidaire found the same thing about leftovers. Three out of four people throw leftovers away every week, and the number one reason was that they forgot. These are industry surveys, not science, so hold them loosely. But they say the same thing your own fridge says.

We misread the date. A study at Ohio State University asked 500 Americans about food waste. Sixty-eight percent believed that throwing food out after the date on the package lowers their chance of getting sick. Here is the thing: it almost never does. Sell by, best if used by, and use by are all about quality. They are the maker's guess at when the food tastes best. Federal law does not require any of them, and none of them is a safety date. The one exception is infant formula.

We buy too much. In the same Ohio State study, 53 percent of people admitted they waste more when they buy in bulk. The big warehouse bag is cheaper per pound right up until you throw half of it away.

And one more number from that study, because it explains why this post exists. Eighty-seven percent of people said they waste less food than similar households. That cannot be true. Somebody in that number is wrong, and it is probably most of us. I am not going to argue with you about it. I am going to show you my fridge.

#### WHY WE THROW AWAY GOOD FOOD

**The date on the package is almost never a safety date.**

##### **SELL BY**

how long the store may display it

##### **BEST IF USED BY**

the maker's guess at peak taste

##### **USE BY**

last date for peak quality

#### **ALL THREE MEAN QUALITY, NOT SAFETY.**

Federal law does not require any of them. The one exception is infant formula.

**68%**

of people believe tossing food after the package date lowers their chance of getting sick. It almost never does.

**77%** feel guilty when they throw food away.

**53%** waste more when they buy in bulk.

##### **How long food really lasts — USDA FoodKeeper**

|                       |                  |         |
|-----------------------|------------------|---------|
| Eggs, raw in shell    | <b>3-5 weeks</b> | fridge  |
| Ground beef           | <b>1-2 days</b>  | fridge  |
| Hard cheese, opened   | <b>3-4 weeks</b> | fridge  |
| Hard cheese, unopened | <b>6 months</b>  | fridge  |
| Chicken, whole        | <b>1 year</b>    | freezer |
| Dry pasta             | <b>2 years</b>   | pantry  |

Sources: USDA FSIS, Food Product Dating · Qi & Roe, PLoS ONE 2016 (Ohio State, 500 U.S. adults) · USDA FSIS FoodKeeper database (658 foods, public domain) — the same table the fridge app uses.

What the date on the package actually means. Chart: Todd Durant. Sources on the chart.

## 6 of 12 · What it costs the planet

I will keep this short, because you have heard it before and because the money argument is the one that changes behavior. But the numbers are real, and they come from the EPA.

Growing the food America throws away uses an area of farmland the size of California and New York put together. It uses enough water and energy to supply more than 50 million homes. When that food rots in a landfill, it gives off methane, and the EPA says wasted food is the source of 58 percent of the methane coming out of America's landfills. Add it all up and the greenhouse gas from wasted food equals more than 42 coal-fired power plants running all year.

Food is also the single most common thing in a landfill. About 24 percent of everything we bury is food.





Wasted food causes 58 percent of the methane from U.S. landfills. U.S. Environmental Protection Agency.



Where the emissions come from, from the farm to the landfill. U.S. Environmental Protection Agency.

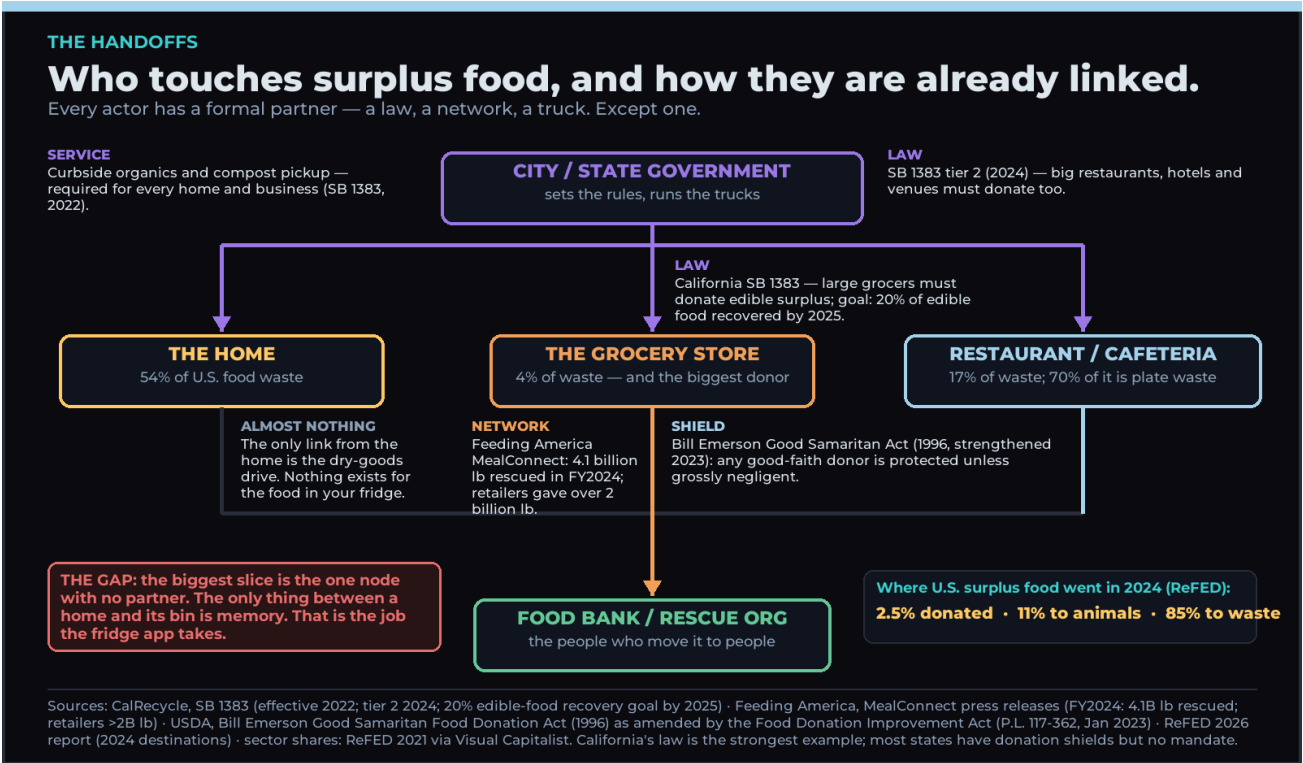
## 7 of 12 • You eat at work, too

Everything above is about your kitchen. But most of us eat somewhere else at least once a day. At work, at school, at a restaurant, at a game. ReFED found that 46 percent of surplus food comes from consumers eating in and out of the home. In restaurants and cafeterias, 70 percent of what gets wasted is plate waste. That is food that made it all the way to a plate and then got scraped into a bin.

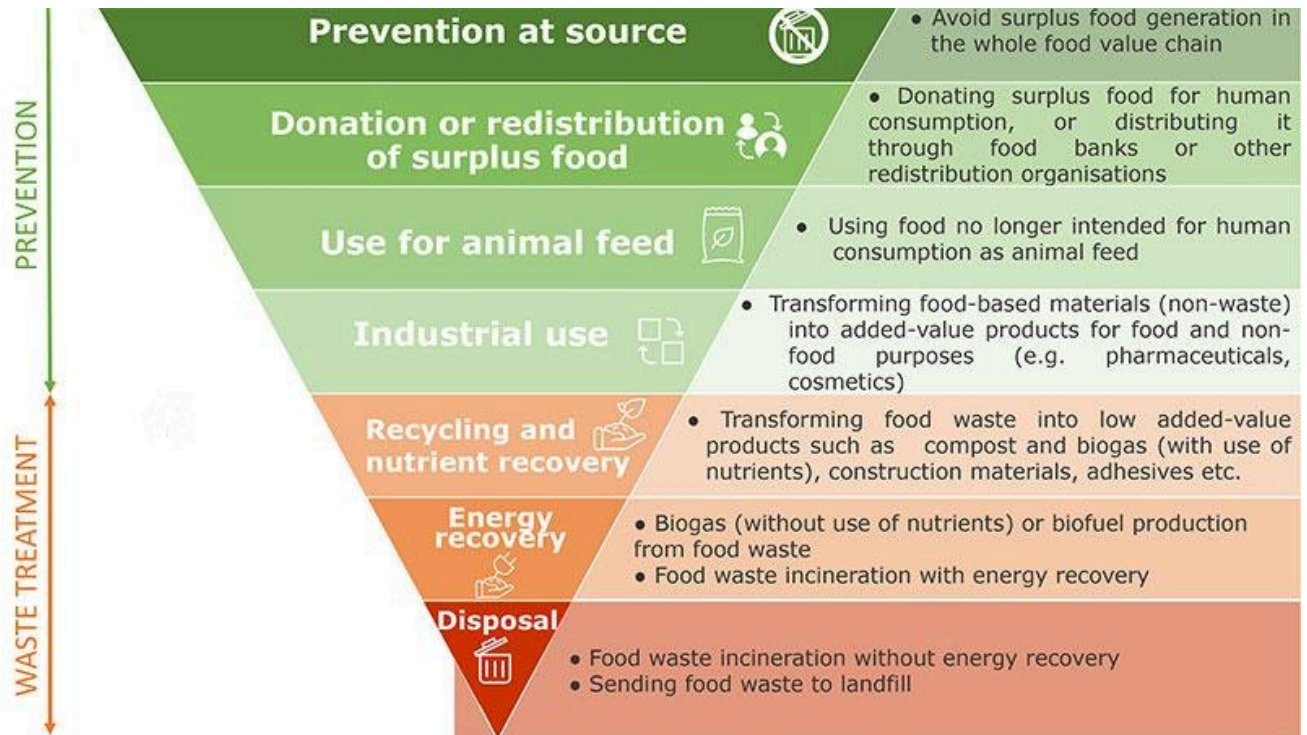
People care about this more than you might think. A Texas A&M researcher surveyed about 1,400 Americans in 2018 and found three out of four were concerned about food waste at organizations, not just at home. In Brisbane, Australia, restaurant owners said they would pay about \$76 a month for a better way to handle their food scraps. The wish is there. The system is not.

So if you run a kitchen, or you order the office lunch, or you pick the caterer for the party, you have a bigger lever than I do. Here is the ladder the European Union uses to decide what to do with food that is not going to be eaten. The EPA has its own version. They agree on the order. Prevent it first. Give it to people second. Feed it to animals third. Compost is near the bottom. The landfill is last. Look at where you are standing on it.

And here is the part that surprised me when I mapped it. The links between these groups already exist, and they are strong. The city sets rules: in California, since 2022, large grocery stores must donate their edible surplus, and by 2025 the state wants a fifth of all edible food recovered instead of dumped. Grocers have a network: Feeding America's MealConnect moved more than four billion pounds of rescued food in one year, and stores are its biggest donors. Restaurants have a shield: a federal law from 1996, strengthened in 2023, protects anyone who donates food in good faith. Every actor in the chain has a partner. Except one. The home. The only thing between your fridge and your bin is memory. That is the gap this whole project lives in.



Who touches surplus food, and how they are already linked. Every node has a partner except the home. Chart: Todd Durant. Sources on the chart.



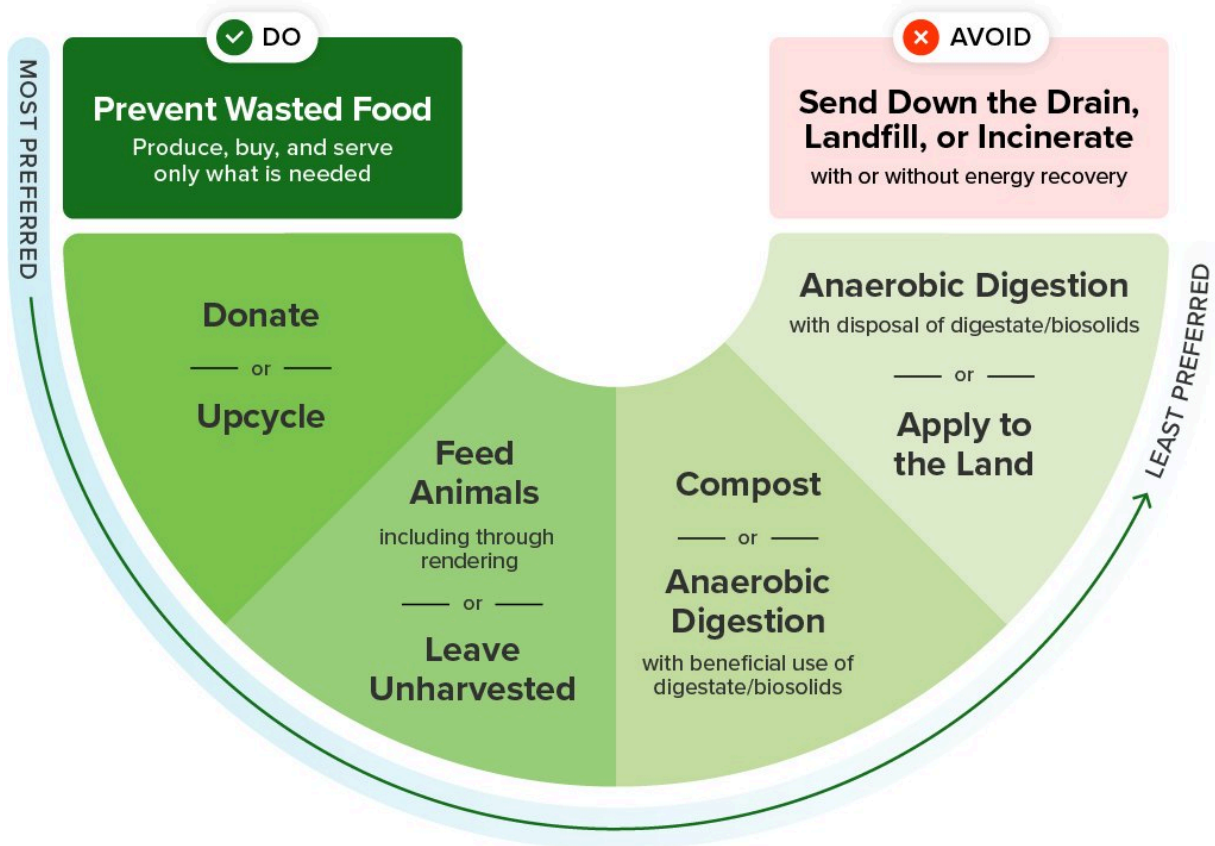
The EU food use and waste hierarchy. Every rung above the landfill is a decision somebody makes. European Commission, Food use and waste hierarchy, 2024. CC BY 4.0.





## Wasted Food Scale

How to reduce the environmental impacts of wasted food



December 2025

The EPA's Wasted Food Scale. Same idea, American version. U.S. Environmental Protection Agency.

## 8 of 12 · How do you actually stop wasting food?

Here is the good news. Small habits work, and there is a study to prove it. The EPA ran a program called Food: Too Good To Waste in cities across the country. Families were given a few simple tools: plan meals, shop with a list, store food where you can see it, eat the oldest things first. After five weeks, the families that stuck with it wasted about a quarter less food. Some cut it by more than a third. No gadgets. Just attention.

Three habits come up in every study, and they cost nothing.

Keep an eat-first bin. One clear box in the middle of the fridge for anything that will go bad in the next two days. When you open the door, that is the first thing you see.

Make food visible. Wash and chop vegetables when you bring them home and put them in clear containers. If you cannot see it, you will forget it.

Freeze instead of tossing. If the meat or the berries have a day left, freeze them. Freezing stops the clock and buys you months.

So why do we still waste food? Because all three habits depend on the one thing the refrigerator never solved. Remembering.



The national goal: cut food waste in half by 2030. U.S. Environmental Protection Agency.

## 9 of 12 • The fridge that remembers

There are apps for this. They ask you to scan your receipts, type in every item, and enter the date it will expire. They work for about a week. Then life gets busy, you stop updating them, and the app becomes a graveyard of food you ate a month ago. The app failed because it added work to your kitchen. If you have to give someone instructions, the design is not done.

So I built one that does not ask. You say what you bought, out loud, as you unpack the bags. It runs inside Home Assistant, the free home-automation system, as a fridge inventory that listens instead of asking. Add milk, chicken, and a bag of pasta. That is the whole job. The system looks each item up in the U.S. Department of Agriculture's FoodKeeper database, which lists how long 658 kinds of food last in the fridge, the freezer, and the pantry. It is free and it is public domain. The pasta goes to the pantry with a two-year clock. The chicken gets two days. Nobody typed a date.

On the fridge door is a small color screen that runs for months on a battery. It shows the list, oldest first. Green means fine. Yellow means eat this soon. Red means you missed it. When you finish something, you say so, from any room in the house. We finished the milk. It is gone from the list.

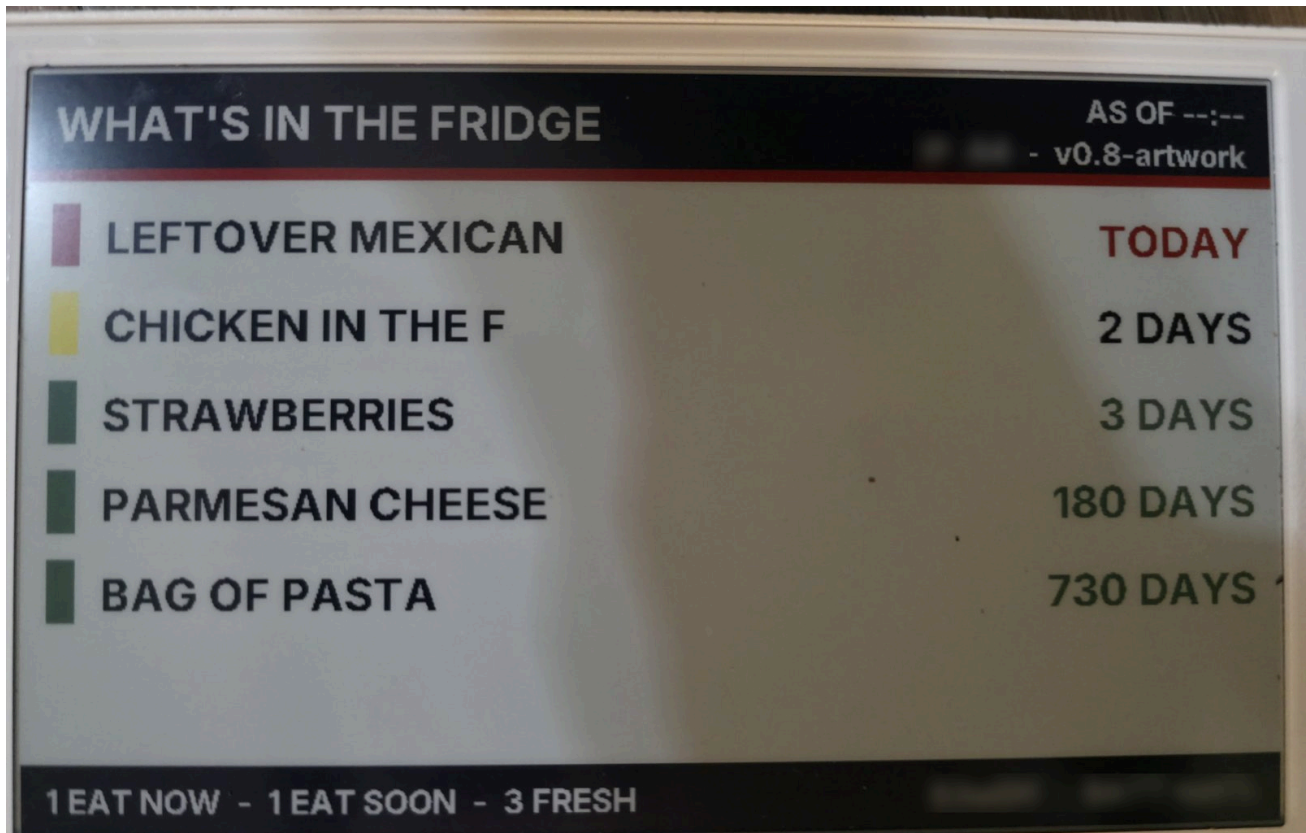
It is not perfect. It will guess wrong sometimes. Fixing a wrong guess takes four seconds. Asking you a question about every item, forever, would take a lot more than that, and you would stop using it. An app that is roughly right about 95 percent of your food tells you more than a barcode scanner that is perfectly right about the 40 percent you bothered to scan.

The build is free. The parts are on the video and the code is on GitHub, linked below.

### LINKS

- Video, the fridge that remembers (episode 1):  
[https://youtu.be/o\\_Ia6XLuVCg](https://youtu.be/o_Ia6XLuVCg)

- Code and build notes: <https://github.com/todddurant-gif/whats-in-the-fridge>
- Help setting it up, by the hour: <https://todddurant.com/consulting>
- Subscribe (RSS): <https://todddurant.com/blog/f.rss>
- Questions and corrections: Hello@todddurant.com



The fridge that remembers: leftover Mexican food is red and due today, the thawed chicken is yellow with two days left, the pasta has two years. Nobody typed a date. Photo: Todd Durant, 2026-09-01. Seed reTerminal E1002 running the Freshness Traffic Light.

## 10 of 12 • The part I need from you

Every human eats. So every human can do something about this, today, without waiting for anyone. That is rare. Most big problems do not give you a lever you can reach from your own kitchen. This one does.



But I only have one fridge, and one family, and I promise you we have not thought of everything. So tell me what I missed. What is in your fridge right now that this would get wrong? What do you do with food that I have not accounted for? And if you run a kitchen, order the office lunch, or work for a city: what is missing from the handoff map? Email me at [Hello@todhdurant.com](mailto:Hello@todhdurant.com), or leave it in the comments under the video. Keep it kind, keep it about the food, and let's make the ideas better together. And if you want help setting this up in your own home, that is what I do by the hour: [todhdurant.com/consulting](http://todhdurant.com/consulting).

Next up: the other half of the answer. Not just wasting less, but growing some of it yourself. That post is coming for the fall planting season.

## **11 of 12 • Quick answers**

### **How much food does the average family waste?**

About \$56 a week for a U.S. household of four, or \$728 per person per year, by the EPA's April 2025 estimate. The EPA says that number is probably low.

### **Is the date on the package a safety date?**

Almost never. Sell by, best if used by, and use by are quality dates. The only federally regulated safety date is on infant formula.

### **How long do eggs last in the fridge?**

Three to five weeks, according to the USDA FoodKeeper database. Ground beef lasts one to two days. Hard cheese, unopened, six months.

### **What share of food waste happens at home?**

More than half, in every place that measures it: 54 percent in the U.S., 53 percent in the EU, 60 percent worldwide.

## **What can I do today?**

Keep an eat-first bin, store food where you can see it, and freeze anything with a day left. Families that did this in the EPA's pilots cut their waste by about a quarter in five weeks.

## **Does a smart fridge inventory need barcodes?**

No. This one takes what you say out loud and looks each food up in the USDA FoodKeeper database for how long it keeps. No scanning, no typing, no dates to enter.

## **12 of 12 · The whole story in one picture**

If you only share one thing from this post, share this. It is every number above, in order, with the source under each one.

# THE FRIDGE THAT FORGETS

12,000 years of keeping food, and the one problem we never solved

**\$56**

a week, in the trash — the average U.S. home of four (EPA 2025)

**>1/2**

of wasted food is thrown away at home — U.S., EU, world, UK

**70M tons**

of U.S. surplus food in 2024 — 29% of the supply (ReFED)

## 1 • WHAT IT COSTS YOU

### WHAT IT COSTS YOU

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**\$14** a week, per person

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Every \$100 of groceries, the average U.S. home.

Sources: U.S. EPA, Estimating the Cost of Food Waste to American Consumers, April 2025 (2023 data) — \$728/person/yr; household of four = per-person × 4; ~11% of food spending; EPA's literature range \$496–\$1,640 normalized to 2023 dollars · ReFED 2026 (\$762/person) · NRDC 2017 (\$1,800/household).

## 2 • THE BEGINNING

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12,000 years of keeping food. Every step got better at keeping. None got better at remembering.



### WHO COULD AFFORD IT

everyone anyone near salt kings cities the middle class almost everyone anyone with a smart speaker

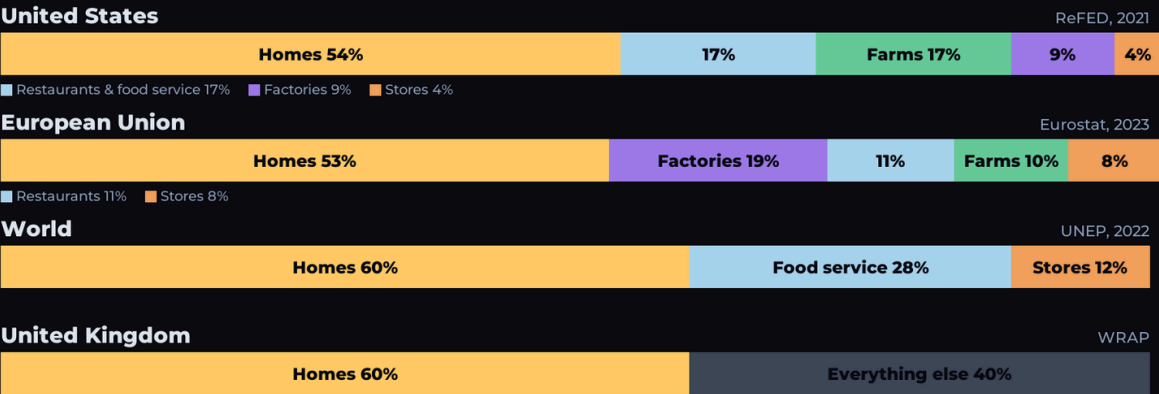
KEEPING got solved seven times. REMEMBERING what is in there, and when it dies, was never on the list until the last step.

Sources: Britannica, food preservation & refrigeration · J. Archaeological Science 2018 (Al Khiday, Sudan: salted fish ~6440 BC) · Lapham's Quarterly 2021 / T. Jackson, Chilled (Shulgi of Ur; Zimri-Lim of Mari ~1780 BC) · ASHRAE (DOMELRE 1913; GE Monitor-Top 1927, >1M sold). Yakhchal date is the commonly cited one.

## 3 • WHERE IT HAPPENS

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More than half of wasted food is thrown away at home — everywhere that measures it.



Four groups, three continents, one answer. And in the U.S., homes are the only sector that shrank in 2024 — nearly a million tons less (ReFED 2026).

Sources: ReFED via Visual Capitalist, U.S. food waste 2021 · Eurostat 2025, EU food waste 2023 · UNEP Food Waste Index 2024, world 2022 · WRAP, UK household food waste. Each dataset defines its sectors slightly differently; all four put homes above half.

4 • WHY WE DO IT

**WHY WE THROW AWAY GOOD FOOD**

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how long the store may display it

**BEST IF USED BY**  
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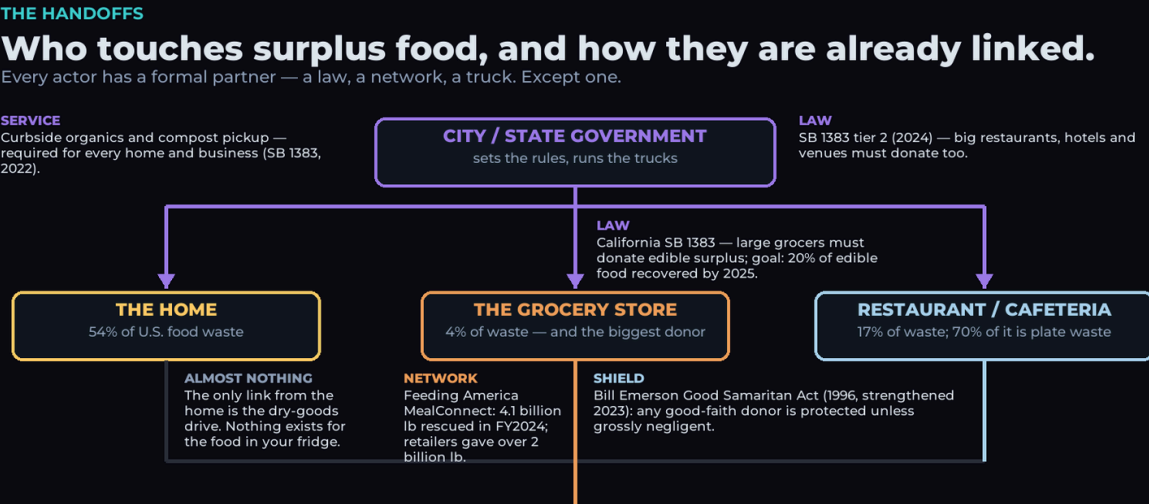
**53%** waste more when they buy in bulk.

**How long food really lasts — USDA FoodKeeper**

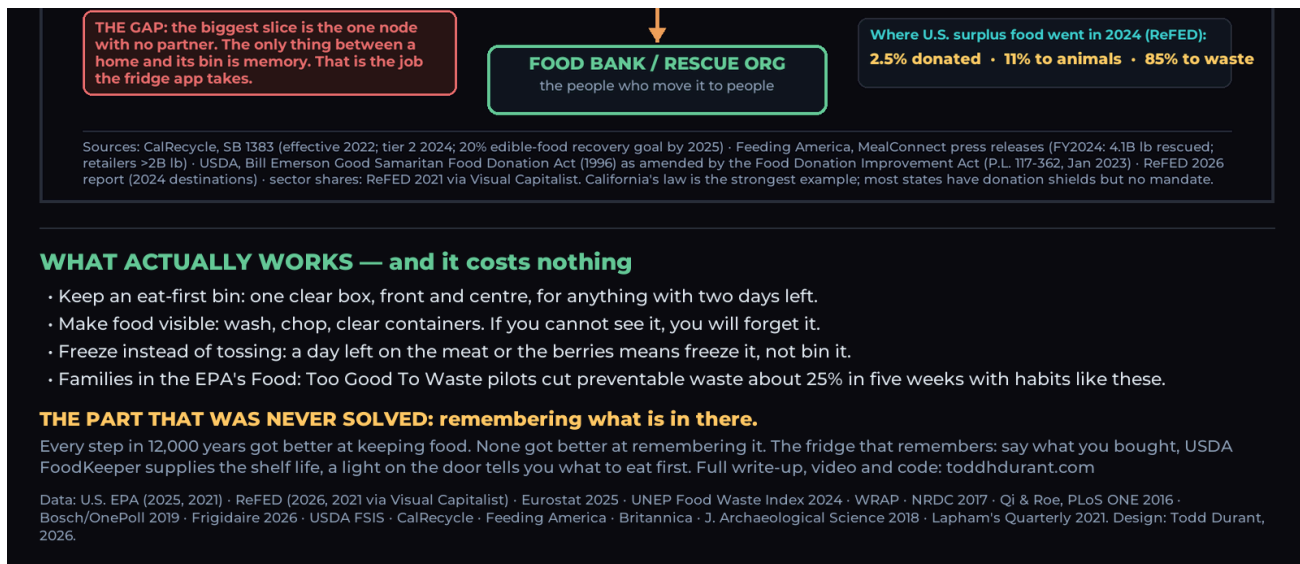
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Sources: USDA FSIS, Food Product Dating · Qi & Roe, PLoS ONE 2016 (Ohio State, 500 U.S. adults) · USDA FSIS FoodKeeper database (658 foods, public domain) — the same table the fridge app uses.

5 • THE HANDOFFS







The Fridge That Forgets: the whole argument in one image. Design: Todd Durant, 2026. Sources on the poster.

## Sources

1. U.S. Environmental Protection Agency. Estimating the Cost of Food Waste to American Consumers. April 4, 2025. Per-person \$728/yr; household of four \$2,913/yr, \$56/wk; ~11% of food spending; 2023 data; literature range \$496-\$1,640 normalized to 2023 dollars. [epa.gov/land-research/estimating-cost-food-waste-american-consumers](https://epa.gov/land-research/estimating-cost-food-waste-american-consumers)
2. ReFED. Progress on the Plate: 2026 U.S. Food Waste Report (2024 data). 70 million tons surplus; 29% of supply; \$380 billion; 412 lb per capita; 46% from consumers in and out of home; residential -950,000 tons. [refed.org](https://refed.org)
3. ReFED. The Problem (web page): 70% of foodservice waste is plate waste; \$762 per person. [refed.org/food-waste/the-problem](https://refed.org/food-waste/the-problem)
4. Visual Capitalist / Selin Oguz. Visualized: Food Waste in the United States. April 8, 2024. Data: ReFED, 2021 (80.7 million tons; residential 54%). [visualcapitalist.com/cp/visualized-food-waste-in-the-united-states](https://visualcapitalist.com/cp/visualized-food-waste-in-the-united-states)
5. Eurostat. Food waste and food waste prevention - estimates (2023 data, extracted September 2025). 58 million tonnes; 130 kg per inhabitant; households 53%.
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